Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	21 October 2026
Team ID	EN25CS3010849
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, SQL HTML, CSS, etc	Used Python (Pandas/NumPy) and SQL scripts to clean, transform, and run statistical aggregations on large Olympic datasets efficient handling of rows across multiple files.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	Snowflake / MySQL / Excel / CSV	Used structured CSV files (medals.csv, athlete.csv, athlete_events.csv) and relational database tables to store processed historical athlete records and medal tallies securely.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Snowflake Stages / COPY INTO / Tableau Data Connections	Ingested raw datasets directly into Python notebooks and BI tools using standard file connectors for seamless data loading.

4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Eg. Tableau prep, Snowflake SQL, Tableau Desktop	Performed null value handling, duplicate removal, text standardization, and created calculated measures (e.g., Total Medals, Medal Success Ratio, Age Groups).
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Snowflake SQl Charts, Tableau , Python	Built interactive dashboards featuring bar charts, line trends, scatter plots, pie charts, and KPI cards to visualize global Olympic performance trends.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, Flask,Streamlit	Provided a clean UI for sports analysts to apply dynamic filters (Country/NOC, Sport, Medal Type) and interact with real-time visual analytics.